Input / Output

General Input / Output

- Ports
  CPUs interface to I/O device
  Specific Memory Address
  Normally a single byte
- Registers
  Control / Status / Data Transfer
- Handshaking
  1. Device raises Ready signal (CTS/RDY)
  2. Data placed on Data Bus
  3. Data Valid (DAV) signal raised
  4. Data read off Data Bus
  5. Data Acknowledged (DAC/ACK) raised
     Possible Timeout on ACK

Memory Mapped I/O

- Specific Known Address
- Memory Read / Write ⇔ Port Input / Output
- Busy Wait or Blocking
  Program has to wait for device
  to be ready for data transfer
- Polling Loop:
  repeat
  Read status register
  until CTS or RDY
  Transfer Data

Interrupt Driven I/O

- Interrupt vector configured / hard-wired
- Device Interrupts CPU when ready

  ⇒ Uses different Interrupts for different functions

  ⇒ Use single interrupt and status register

  ⇒ Daisy-Chaining
- Non-blocking
  CPU continues with another task while it waits for the device to raise an interrupt request for data transfer

Direct Memory Access (DMA)

- Used for large data transfers
- Uses the Address and Data Buses independently of the CPU
- DMA Controller (DMAC):
  Four configuration Registers

Start Address	Location in memory
Port Address	Location of device
Byte Count	Size of data transfer
Direction	To or From memory

  Controller interrupts CPU when transfer complete

DMA Modes

- **Burst** Mode:

  Suspends CPU memory access

  Extremely fast data transfer
- **Cycle Stealing** Mode:

  Interleaved with normal memory access

  DMAC access memory when CPU is busy

  Slower than Bust, much faster than program control

  Competes for memory cycles with RISC processors

Interfaces

- **Parallel** Interface
  Transfer one byte / word at a time
  Fast – measured in Kilo- or Mega-Bits per second
  Costly – Duplicate hardware
  Prone to error over large distances

- **Serial** Interface
  Transfer one bit at a time
  Slow – measured in Kilo-Bits per second (baud)
  Two sides synchronise with a *Start* bit
  Uses *Parity* bit as simple error check
  Cheep – Almost no hardware required
  Can cover very large distances

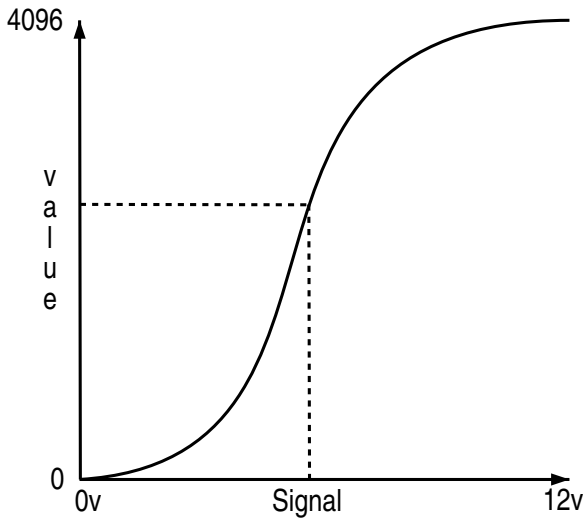
Analog Interface

- Computer are Digital !
- The 'Real World' is Analog !

- **ADC**: Analog to Digital Converter
  Uses analog signal to return a ratio
  between two values, normally 0 . . . 4K
  ADC must be calibrated for each job
  Not exact, as value is an estimate

- **DAC**: Digital to Analog Converter
  Uses digital value (between 0 and 4K) as
  ratio between upper and lower voltage limits

ADC Curve



Example I/O Devices

Input
Mouse
Sensor
Scanner
Keyboard
Microphone
Microswitch

Output
Motor
Printer
Speaker
TV / CRT
Video Display
Liquid Crystal Display

Storage
Tape
Disk
Hard Disk
Optical Disk

Types of Printer	
Laser	Cylinder
Ink-Jet	Golf-Ball
Bubble-Jet	Line / Belt
Dot Matrix	Daisy-Wheel